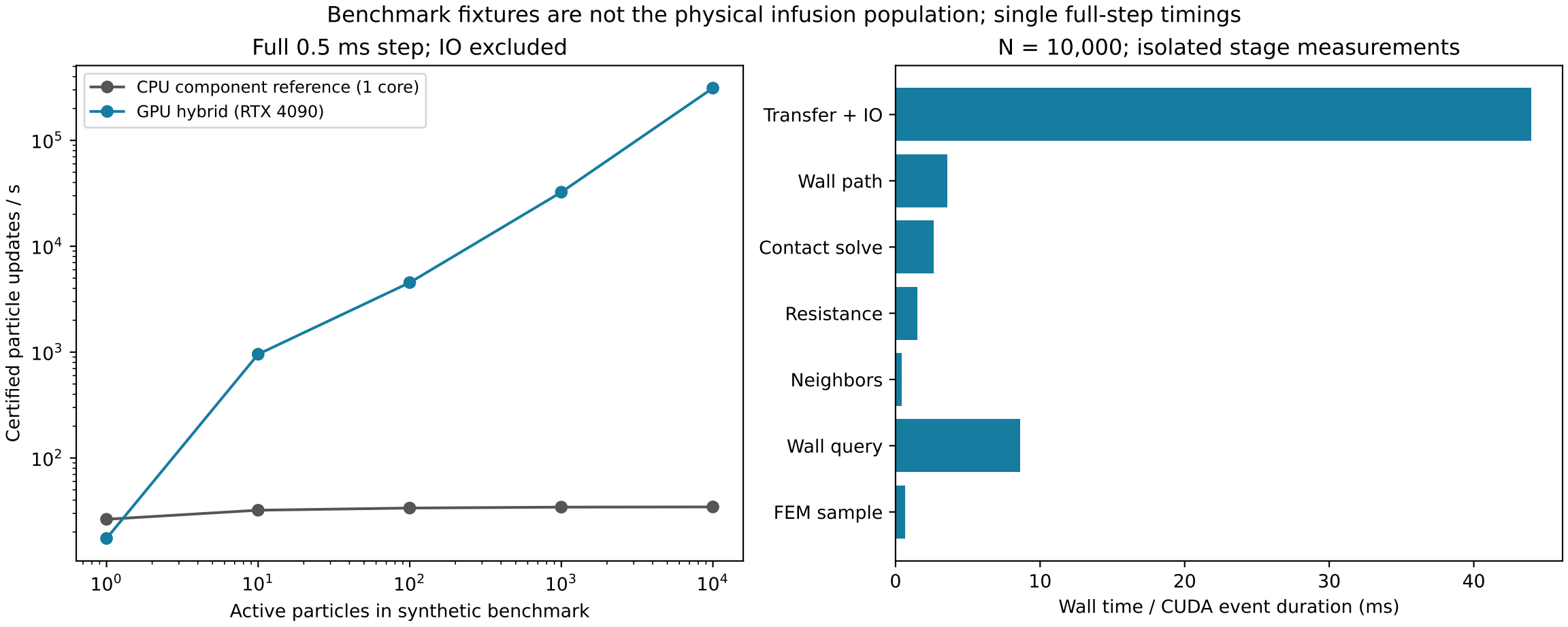
Benchmark fixtures are not the physical infusion population; single full-step timings
Full 0.5 ms step; IO excluded
CPU component reference (1 core)
GPU hybrid (RTX 4090)
Certified particle updates / s
10^5
10^4
10^3
10^2
10^0
10^1
10^2
10^3
10^4
Active particles in synthetic benchmark
N = 10,000; isolated stage measurements
Transfer + IO
Wall path
Contact solve
Resistance
Neighbors
Wall query
FEM sample
0
10
20
30
40
Wall time / CUDA event duration (ms)